Doriane **Ouédraogo**

+1 (514) 519-8940 | ioued078@uottawa.ca | Doriane Ouédraogo | GitHub | Gatineau, Québec | Languages: French | English

Profile

Mechanical Engineering and Computing Technology student at the University of Ottawa, passionate about the design and optimization of mechanical and mechatronic systems. Experience in dynamic modeling, complex assembly design, and control system development under technical constraints.

Technical Skills

Design & Simulation: SolidWorks, AutoCAD, SAP2000 | **Control & Programming:** MATLAB, Python, Arduino, PID, VBA **Manufacturing:** CNC machining, conventional lathe, 3D printing, laser cutting | **Analysis & Optimization:** Lean methods, statistical analysis, quality control | **Technical Tools:** Trello, Jira, Notion, Excel (macros), Microsoft Access

Academic projects and achievements

Robotic Coil Winding Assembly – Composite Manufacturing 3DTO

(SolidWorks / Trello / Miro)

University of Ottawa
(January 2026 – April 2026)

- Designed a controlled carbon fiber winding system under misalignment constraints (±10 mm / 10°)
- Developed a tension-control and twist-elimination mechanism
- Produced complete technical documentation for manufacturing and validation

Inverted Pendulum on Linear Cart

(SolidWorks / PID / 3D Printing / Arduino)

University of Ottawa
(January 2026 – April 2026)

- Designed a mechatronic system stabilized using a PID controller on Arduino
- Linearized and experimentally validated a nonlinear dynamic system
- Optimized dynamic response and reduced oscillations

Dynamic Modeling of an Active Race Car Suspension System

(SolidWorks / MATLAB)

University of Ottawa
(September 2025 – December 2025)

- Modeled and simulated the dynamic response of a suspension system
- Analyzed the influence of control parameters on system stability

Paralympic Tricycle Differential System

(SolidWorks / Notion / CNC Milling / Welding / Lathe)

University of Ottawa
(September 2025 – December 2025)

- Contributed to the design, machining, and technical analysis of the system
- Structured the technical and economic documentation
- Project awarded Honourable Mention – Accessibility Category at Design Day

Education and training

- **B.Sc.A. in Mechanical Engineering & Computing Technology- University of Ottawa** — September 2022 – June 2027
- **ASP Construction Certificate** — June 2025
  ASP Construction, Québec
- **Machining Techniques (Lathe) Certificate** — October 2025
  University of Ottawa, Ottawa, Ontario
- **Mechanical Design & Problem Solving (15 Projects)** — December 2025
  Udemy

Professional experience and community involvement

Mentor – Technovation — Since january 2026

University of Ottawa, Ottawa, Ontario

- Mentoring a junior team in the development of a STEM project

Tuteur – Tutorat Pro — Since may 2025

Gatineau, Québec, Canada

- Providing mathematics and science tutoring to improve students' academic understanding and performance

Student involvement :

- Member of BSLA, NSBE, Crochet Club, Cooking Club, Avid Readers Club (University of 'Ottawa).